A Comparative Analysis of File System Type and Digital Forensic Data Recoverability

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***Abstract*—Digital forensic investigations often require recovering deleted or reformatted data from storage media. Existing research has examined file-carving techniques, recovery tools, and individual file systems. However, comparatively less research has directly evaluated how different file-system architectures affect data recoverability under identical experimental conditions. This study proposes a controlled, reproducible comparison of forensic file recovery across NTFS, exFAT, ext4, Btrfs, and bcachefs using a standardized NixOS virtual-machine environment.**

This study will experimentally evaluate whether file-system architecture affects forensic data recoverability under controlled conditions. A reproducible NixOS virtual-machine environment will be used to test NTFS, exFAT, ext4, Btrfs, and bcachefs using fresh virtual disks for each experimental run. A standardized dataset containing known files will be written to each file system and cryptographic hashes will be recorded before a controlled data-loss procedure is performed and file recovery will then be attempted using PhotoRec.

This study aims to determine whether file-system type produces measurable differences in forensic data recoverability when hardware-related factors are intentionally controlled. The resulting virtual test environment is intended to provide a reproducible baseline for comparing file-system behaviour.

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

A. Background and Context

Even after users delete files, format storage media, or lose access to data, digital storage devices often retain substantial residual information that remains valuable for forensic investigations [**?**]. Sibe and Idigbo [**?**] highlight real-world consequences of persistent residual data by examining refurbished hard drives that had undergone formatting and sanitization, successfully retrieving 28,691 files which included sensitive personally identifiable information.

File carving is used when file-system metadata is corrupted, incomplete, or missing, as it reconstructs file content directly from unallocated space using recognized file signatures and internal structural patterns. Tam and Abdullah [**?**] highlight PhotoRec as an effective carving utility that functions independently of active file-system structures and supports numerous file formats. In an experimental evaluation within the CAINE forensic environment, Carranza and DeCusatis [**?**] demonstrated that PhotoRec yields consistent, repeatable recovery performance, reinforcing its suitability as a standardized open-source carving tool. Further empirical studies confirm that PhotoRec can effectively carve target storage partitions without requiring a forensic image. For instance, Pratama [**?**] applied PhotoRec directly to a FAT32 USB flash drive and successfully recovered all 2,781 test files, while Dasmen et al. [**?**] demonstrated the successful retrieval of diverse file types from external flash media. While these findings do not guarantee universal file recovery across all scenarios, they confirm PhotoRec's utility as a reliable signature-based recovery tool.

Nevertheless, recovery success is heavily influenced by the specific file system and operational environment under investigation. Researching NTFS, Venkatesh et al. [**?**] achieved an approximate 95% recovery rate by parsing Master File Table (MFT) records to restore deleted files. In Linux environments, Kim et al. [**?**] demonstrated that alterations to Ext4 journal structures and XFS metadata directly affect tool performance and hash integrity during file reconstruction. Btrfs introduces unique challenges, as Pandey et al. [**?**] note its copy-on-write (CoW) design, subvolumes, and historical B-tree roots preserve recoverable data states but require distinct reconstruction methodologies compared to traditional file systems. Collectively, these insights show that file-system architecture is a key aspect of forensic recoverability and should be evaluated as an explicit experimental variable.

Rather than directly comparing underlying file systems, many other studies focus on specialized embedded platforms. For example, Raymer et al. [**?**] established a standardized forensic acquisition and analysis protocol for the Meta Quest 2, whereas Marques et al. [**?**] investigated QNX6FS on BMW infotainment systems, utilizing PhotoRec to carve unallocated space. Although these investigations emphasize the importance of rigorous, reproducible methodologies, they also illustrate how current literature remains largely tied to specific hardware devices, specialized operating systems, or isolated file-system implementations.

B. Problem Statement

File recovery and file carving are established areas of digital forensics, but recovery outcomes may depend not only on the recovery tool but also on the underlying file system. File systems use different structures to allocate data, record metadata, track deletion, and manage free space. In NTFS, for example, deleted file entries may remain represented in the MFT while

their data persists until the associated storage space is reused [**?**]. Ext4 can retain deleted-file metadata within its journal [**?**], whereas Btrfs uses CoW B-trees and historical filesystem roots rather than traditional fixed metadata structures [**?**]. These architectural differences create a reasonable basis for expecting file-system type to affect what remains recoverable after deletion.

Existing studies provide useful evidence about individual recovery environments, but they generally do not isolate filesystem type under the same recovery environment. Chhajed [**?**] is particularly relevant because the study compares deleted-file persistence across several file systems on both HDD and SSD storage, however, it also varies by operating systems, disk types, and ordinary user activities, making it difficult to attribute observed differences to filesystem architecture alone. Kumar and Haribabu [**?**] hold a Linux partition constant and compare three carving tools against deleted Docker-container data, while Venkatesh et al. [**?**], Kim et al. [**?**], and Pandey et al. [**?**] focus on filesystem-specific recovery techniques. The literature therefore demonstrates that both recovery method and filesystem structure matter, but leaves room for a controlled experiment in which the recovery tool and virtual environment are held constant while only the filesystem is changed.

This study addresses this problem by creating a controlled virtual test environment in which NTFS, exFAT, ext4, Btrfs, and bcachefs undergo the same data-generation, deletion, and recovery procedures. By using fresh virtual disks for each trial, the study seeks to establish a reproducible software-level baseline of file-system recoverability without introducing hardware-specific behaviour such as physical media wear or storage-controller effects.

C. Literature Overview

Existing work relevant to this study can be grouped into four broad themes. First, researchers have evaluated forensic tools and their performance. Carranza and DeCusatis [**?**] experimentally validated open-source tools including PhotoRec within CAINE, while Liambas and Manios [**?**] compared PhotoRec and other forensic utilities across multiple operating systems and repeated each tool test, ten times to obtain the average execution times. Pratama [**?**] and Dasmen et al. [**?**] both used PhotoRec for direct recovery from removable storage. Kumar and Haribabu [**?**] compared Scalpel, Foremost, and PhotoRec when carving data from deleted Docker containers, providing a useful precedent for controlled recovery experiments conducted in virtualized or software-defined environments.

Second, several studies investigate file recovery within a particular file system or storage architecture. Venkatesh et al. [**?**] developed a Python/PyTSK3 approach for recovering deleted NTFS files through MFT analysis. Kim et al. [**?**] developed a TSK-based framework that uses Ext4 journal metadata for deleted-file recovery and XFS metadata for extraction, showing that forensic tool support must adapt when filesystem metadata structures change. Choi and Lee [**?**] similarly combine NTFS metadata analysis with file carving to recover data from damaged RAID systems. Most directly relevant to the present filesystem set, Pandey et al. [**?**] show that Btrfs recovery is shaped by CoW B-trees, historical roots, subvolumes, and that a filesystem-aware method can reconstruct the file content, paths, and metadata when the required structures survive.

Third, file-carving studies have emphasized content-based recovery and integrity verification. Tam and Abdullah [**?**] describe file carving as recovery from raw or unallocated space without depending on active file-system metadata, and their Android JPEG recovery workflow uses hash values to verify recovered-file integrity. Kumar and Haribabu [**?**] found that their carving tools did not preserve original filenames or timestamps and that recovery success varied by file type in their Docker dataset. Pandey et al. [**?**] also distinguish signature-based carving from filesystem-aware reconstruction, noting that carving generally provides weaker path, metadata, fragmentation, and filesystem-consistency guarantees. These findings support the present study's decision to identify recovered artifacts by cryptographic hash rather than filename and to treat exact content integrity as a primary outcome.

Finally, research on real storage devices demonstrates why environmental control is necessary. Sibe and Idigbo [**?**] recovered substantial residual data from refurbished hard drives that had reportedly been formatted and sanitized. Chhajed [**?**] found that deleted-file persistence can be affected by the filesystem, storage medium, operating system, and subsequent user activity. These studies provide strong real-world motivation, but they also illustrate the confounding factors introduced by physical HDDs and SSDs. The present study intentionally removes those hardware variables so that filesystem-dependent differences in Photorec recovery can be examined against a common virtual baseline.

D. Research Gap

Taken together, the initial literature establishes that deleted-data recovery, file carving, forensic-tool performance, and file-system-specific recovery are active research areas. However, the studies reviewed so far generally examine a single file system, a particular device or storage medium, or differences between recovery tools and operating environments. None of the articles in this first set directly compare several modern file systems under the same virtual hardware, test corpus, deletion procedure, and PhotoRec recovery process.

The proposed study therefore shifts the file system itself to the primary independent variable. By testing NTFS, exFAT, ext4, Btrfs, and bcachefs in fresh, standardized virtual disks. The experiment is designed to reduce hardware-related confounders and isolate differences in logical recoverability.

The research gap addressed in this proposal is therefore the limited availability of controlled and repeatable experiments that compare forensic file recovery across multiple file-system architectures while keeping the recovery tool and experimental environment constant.

E. Research Objectives

The objective of this research is to deploy an automated, repeatable virtual testing environment to compare forensic data-recovery outcomes across different file systems while minimizing variation from hardware and experimental setup. This study aims to explore how the underlying filesystem architecture (NTFS, Ext4, exFAT, Btrfs, and bcachefs) affects the subsequent file carving through PhotoRec.

II. Literature Review

Write your previous work here. We explore related work in three broad areas: theme 1, theme 2, and theme 3, followed by a literature review matrix that highlights the unanswered research gaps in each theme.

A. Literature Review Matrix

B. Theme 1

C. Theme 2

D. Theme 3

III. Methodology

Explain the comprehensive approach you used to conduct this research here.

A. Research Method

B. Data Sampling

C. Data Analysis Techniques

D. Data Measurement Models

IV. Evaluation

A. Research Questions/Hypothesis

B. Evaluation metrics

- Metric 1
- Metric 2
- Metric 3
- .....

C. Result

Provide the result of your analysis in the form of figures and tables here and explain how your results answer your research questions

V. Discussion

What challenge did you face? Is your result generalizable? How did you validate your result?

VI. Conclusion and Future Work

What are the existing gaps and unanswered research questions? what is the next step?

Acknowledgment

References

Provide your citations as long as you continue with your research. For example, Einsteign et.al. [**?**] demonstrated that energy converts...

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